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(54) RESILIENT SHEET STRUCTURE AND VOICE COIL MOTOR

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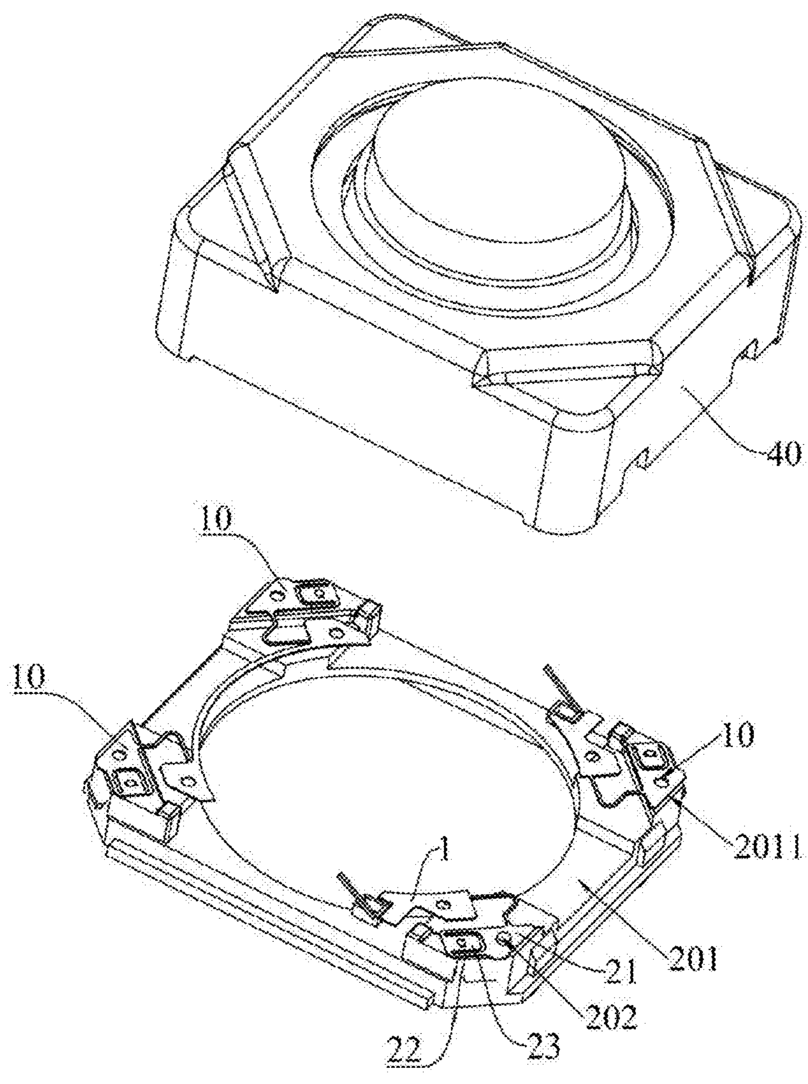
CPC ***H02K 33/02*** (2013.01); *H02K 2203/09* (2013.01)

(57) **ABSTRACT**

Provided are a resilient sheet structure and a voice coil motor. The resilient sheet structure includes a connection piece and a resilient sheet. The resilient sheet includes a fixed portion, a beam, and a fitting portion. The fixed portion is electrically connected to the connection piece and is configured to be connected to the base assembly of a voice coil motor. One end of the beam is connected to the fixed portion, and the other end of the beam can rotate around one end of the beam to enable the beam to be at an included angle with a preset plane. The fitting portion is connected to the other end of the beam, and the fitting portion can rotate around the beam to enable the fitting portion to be parallel to the preset plane.



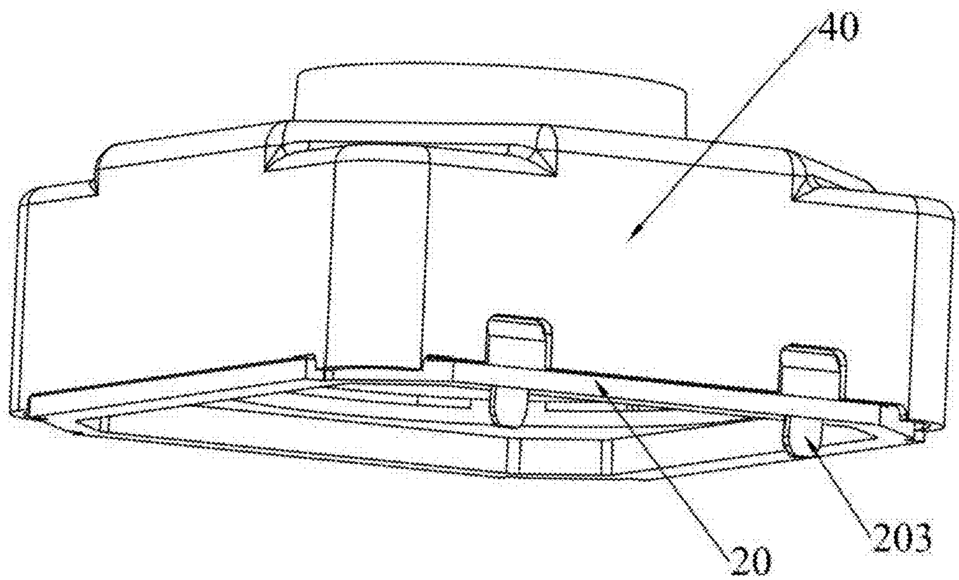


FIG. 1

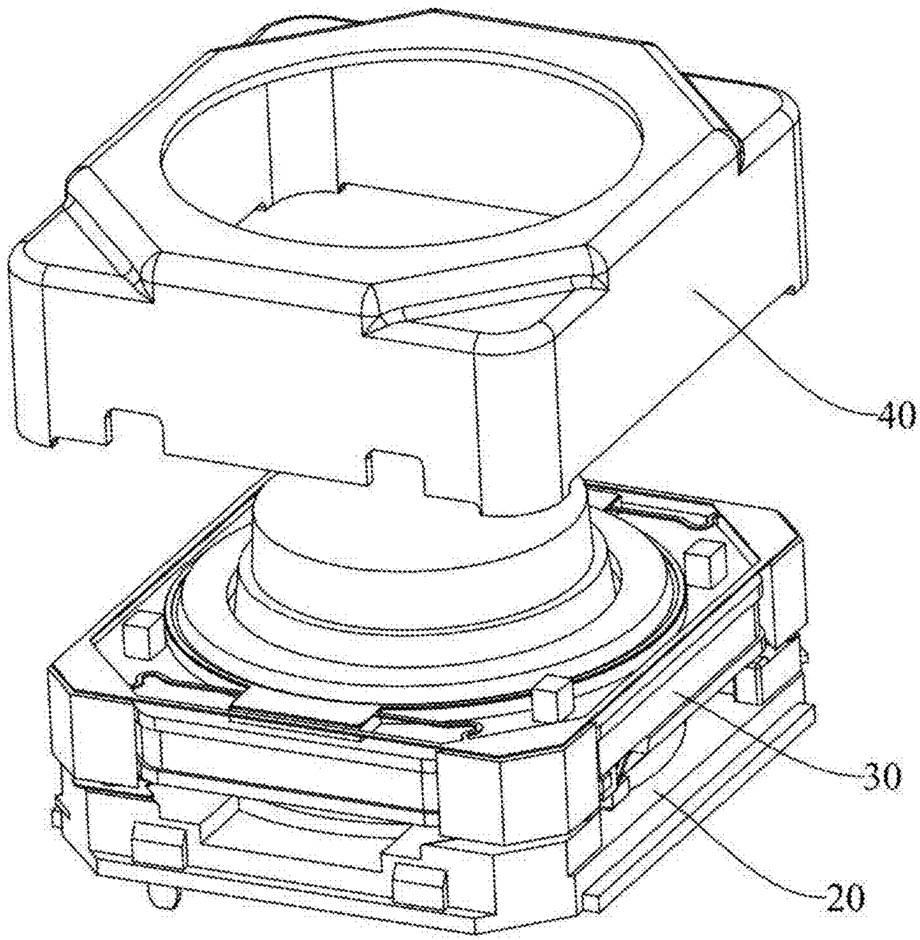


FIG. 2

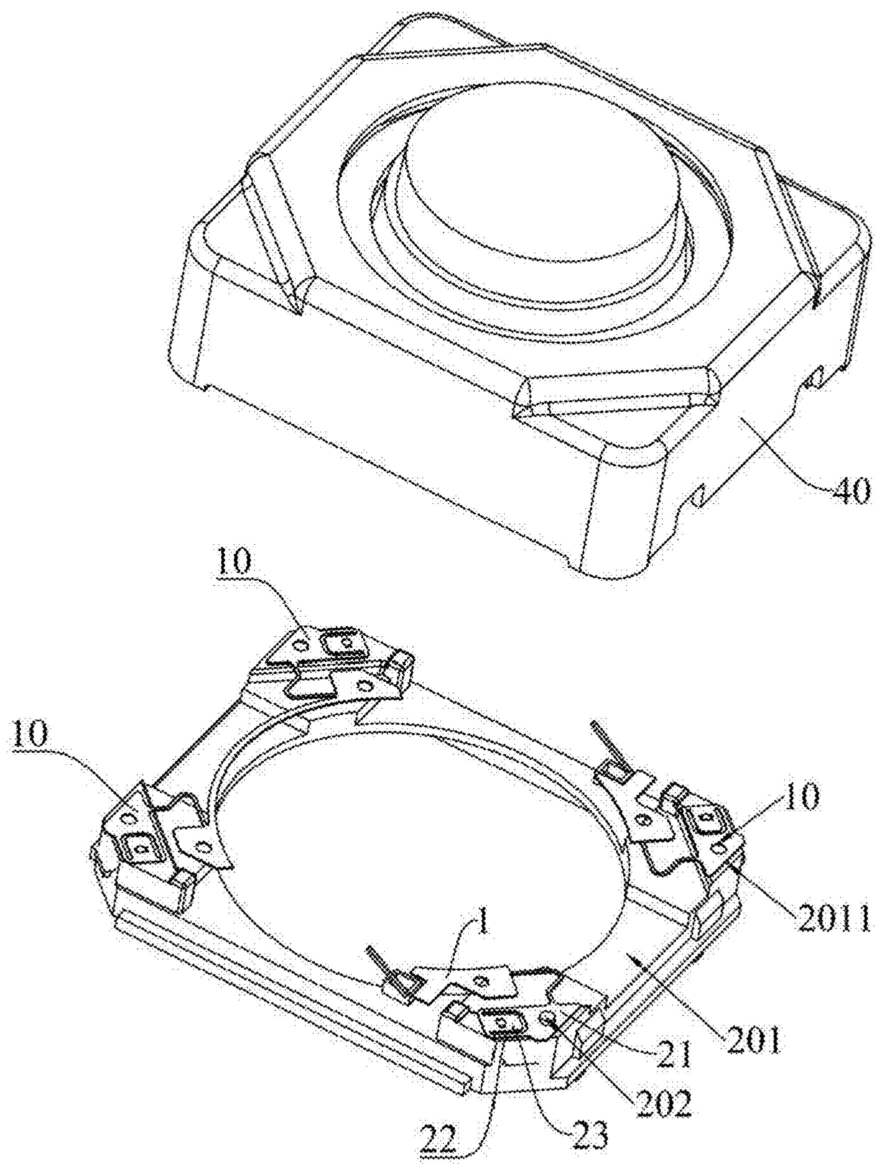


FIG. 3

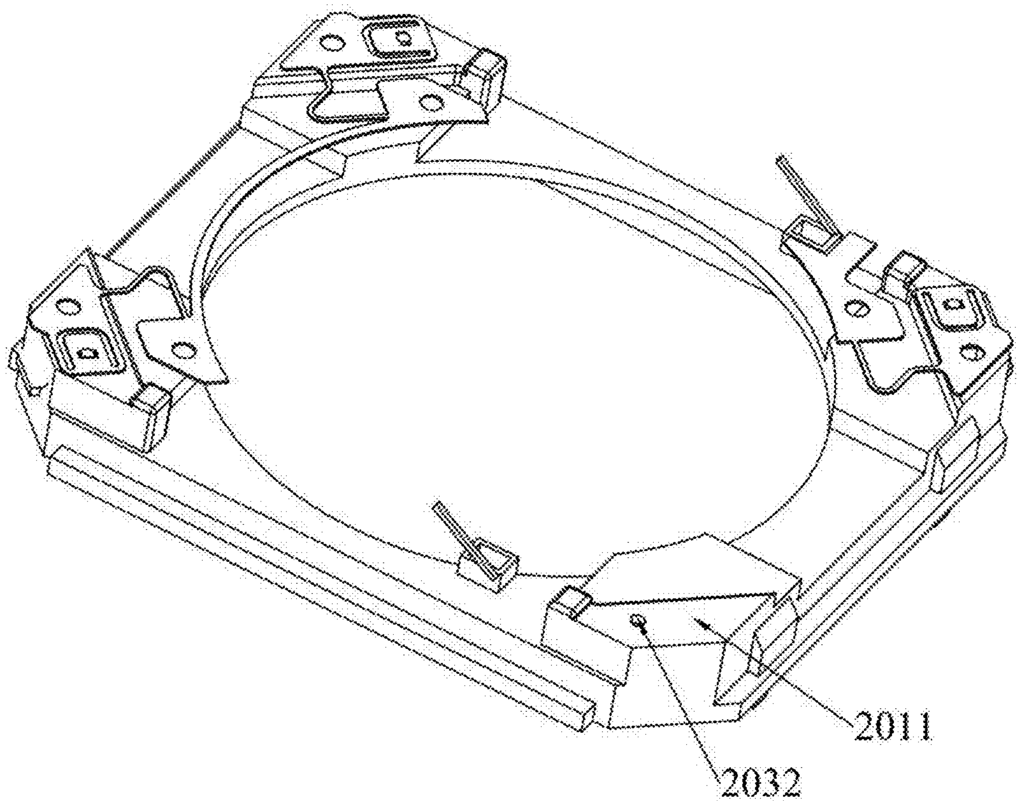


FIG. 4

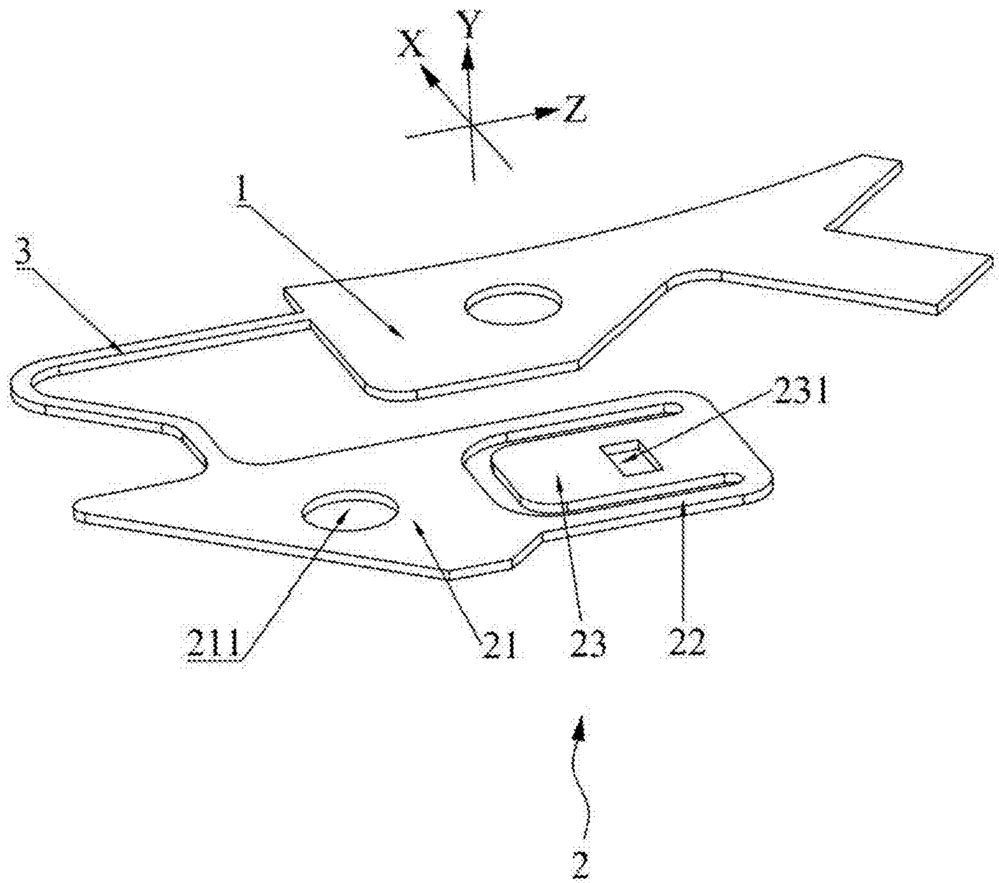


FIG 5

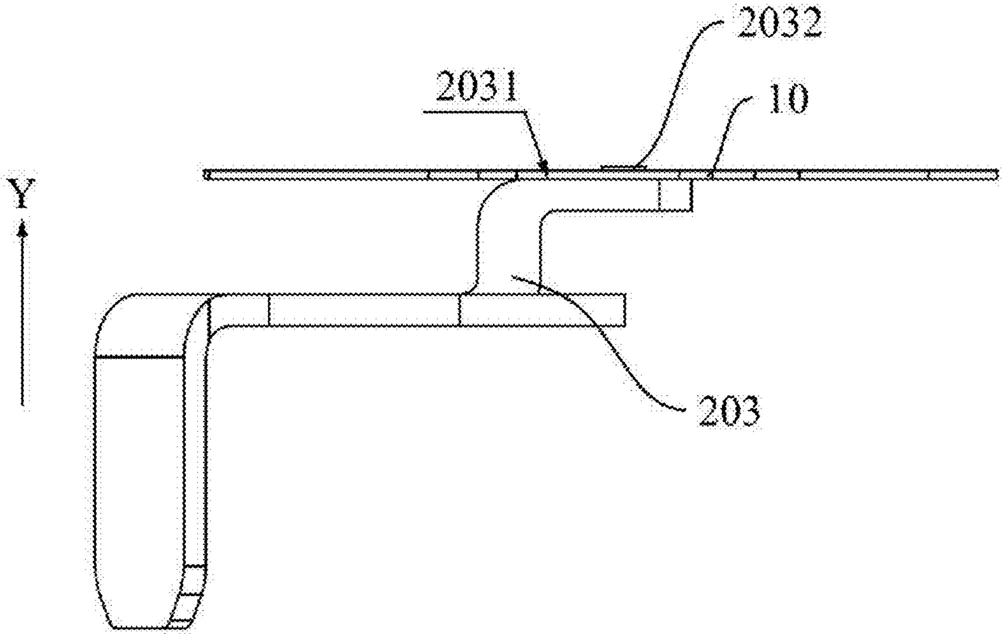


FIG 6

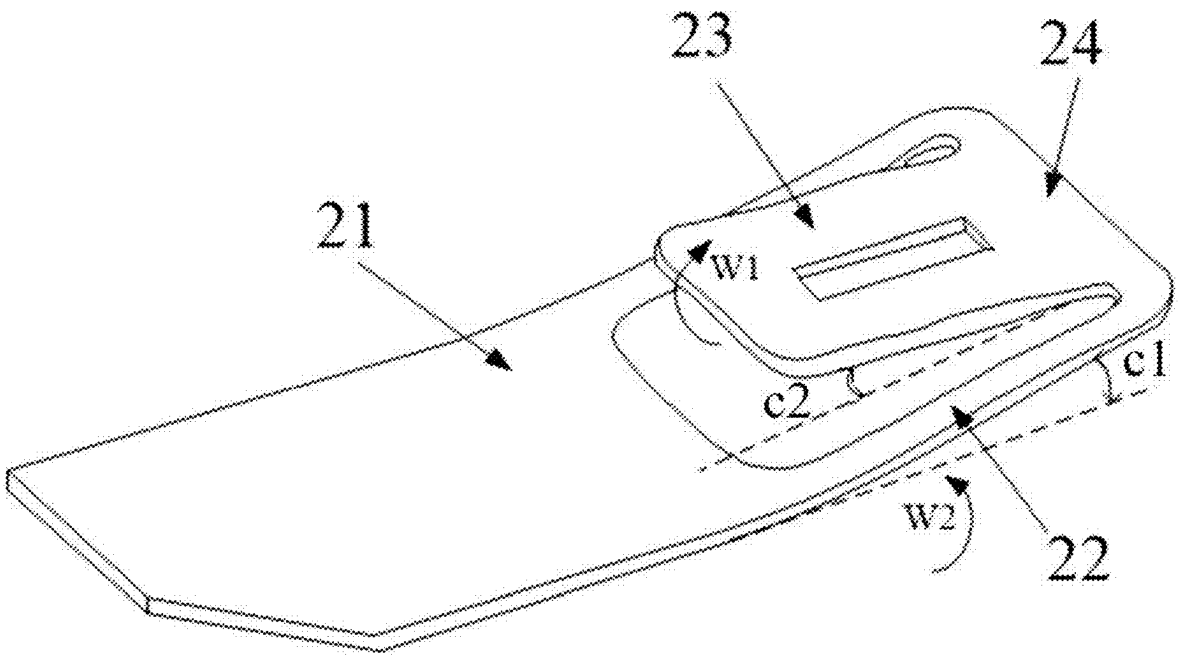


FIG. 7

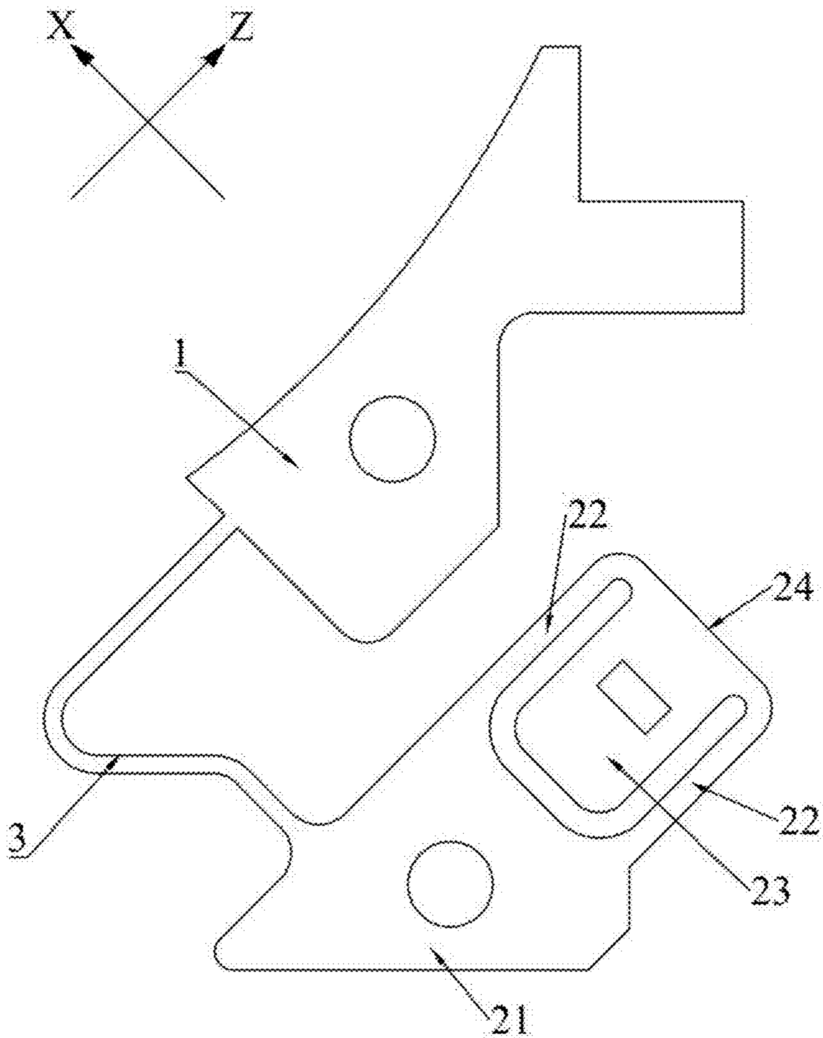


FIG. 8

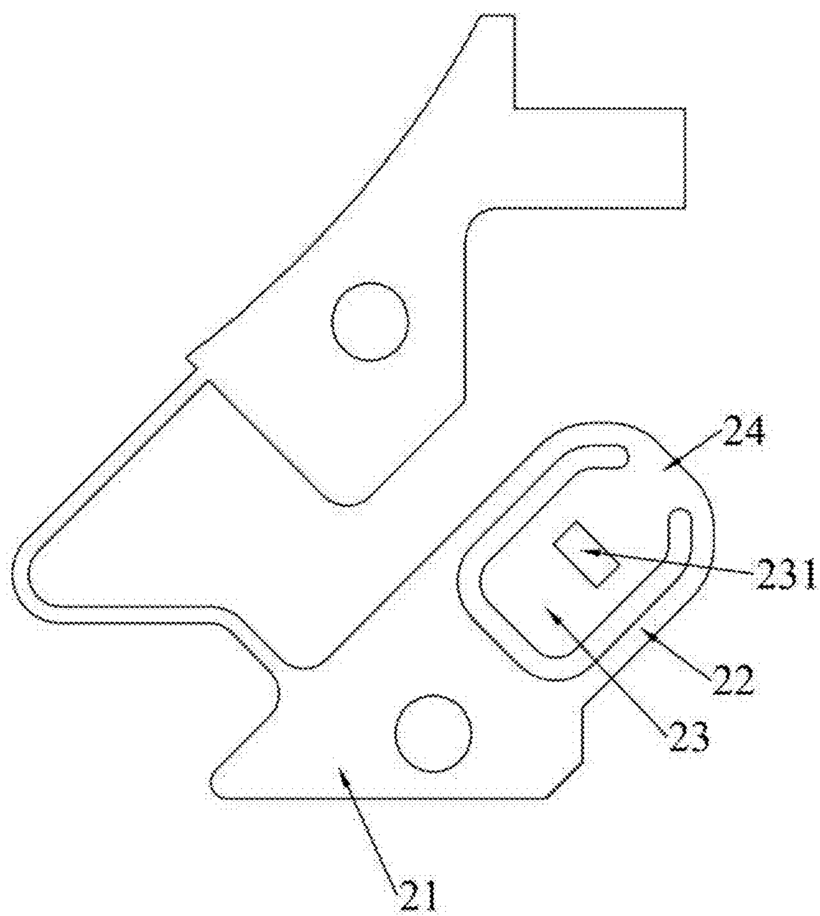


FIG. 9

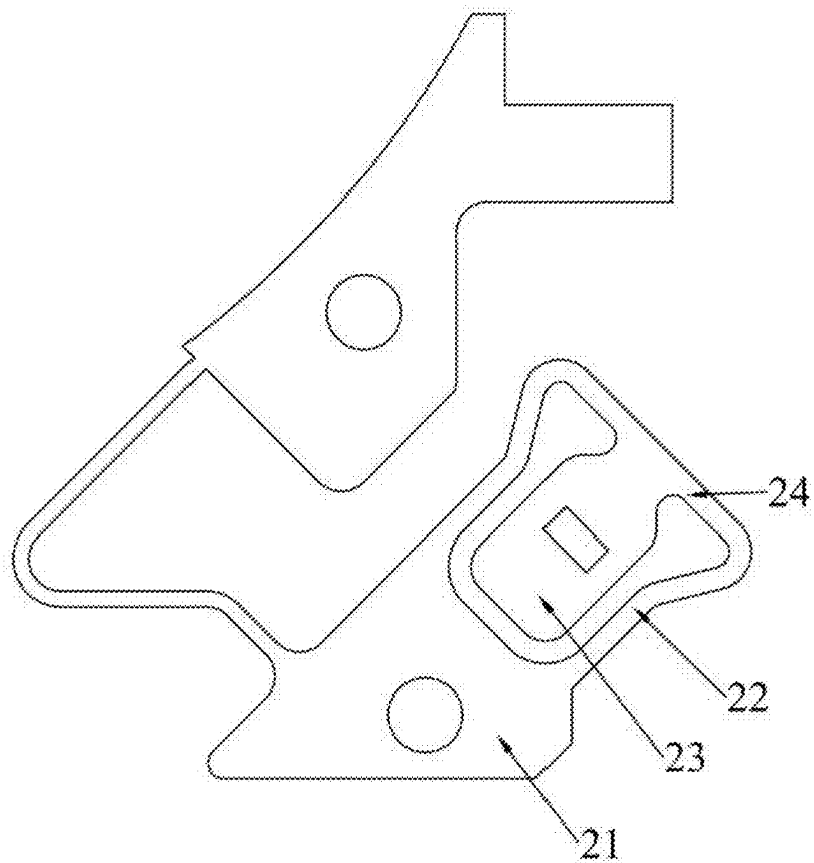


FIG. 10

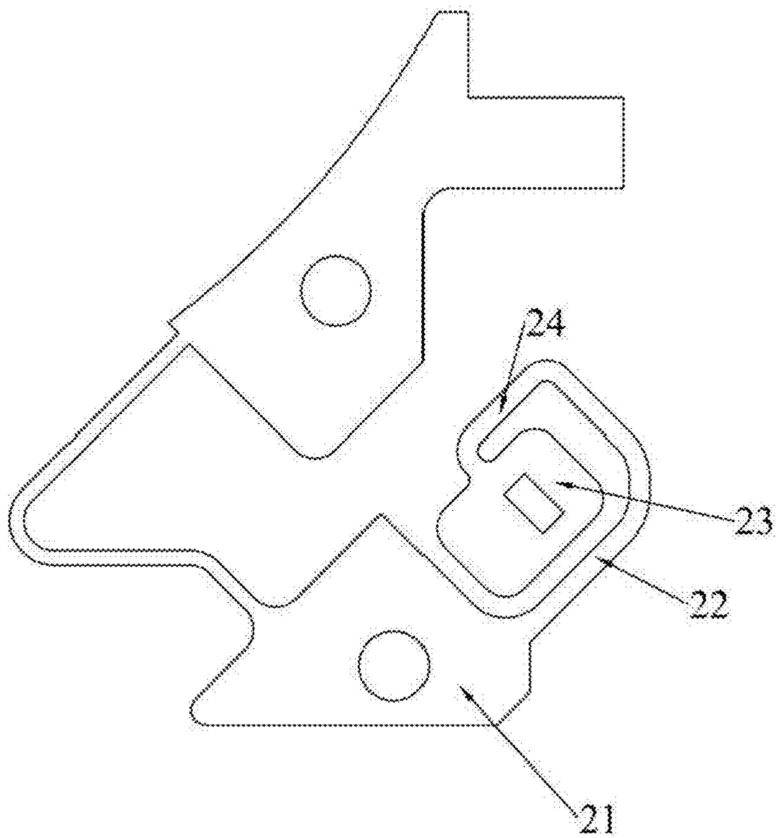


FIG. 11

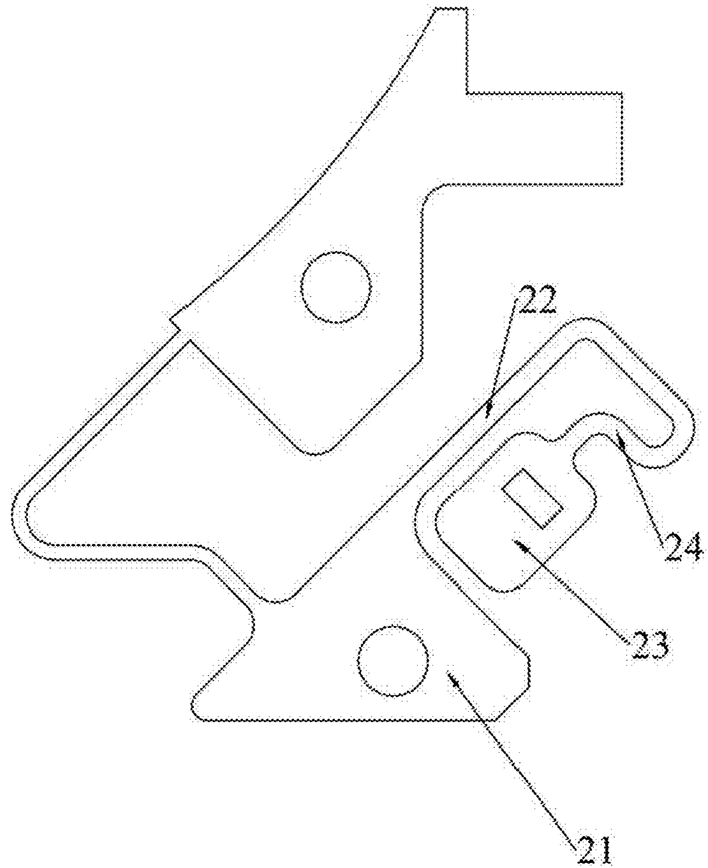


FIG. 12

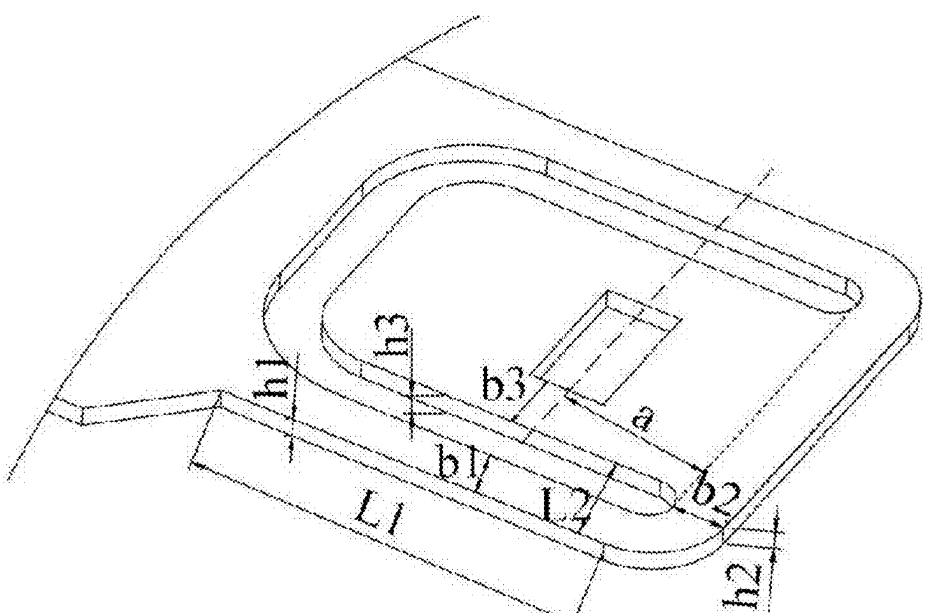


FIG. 13

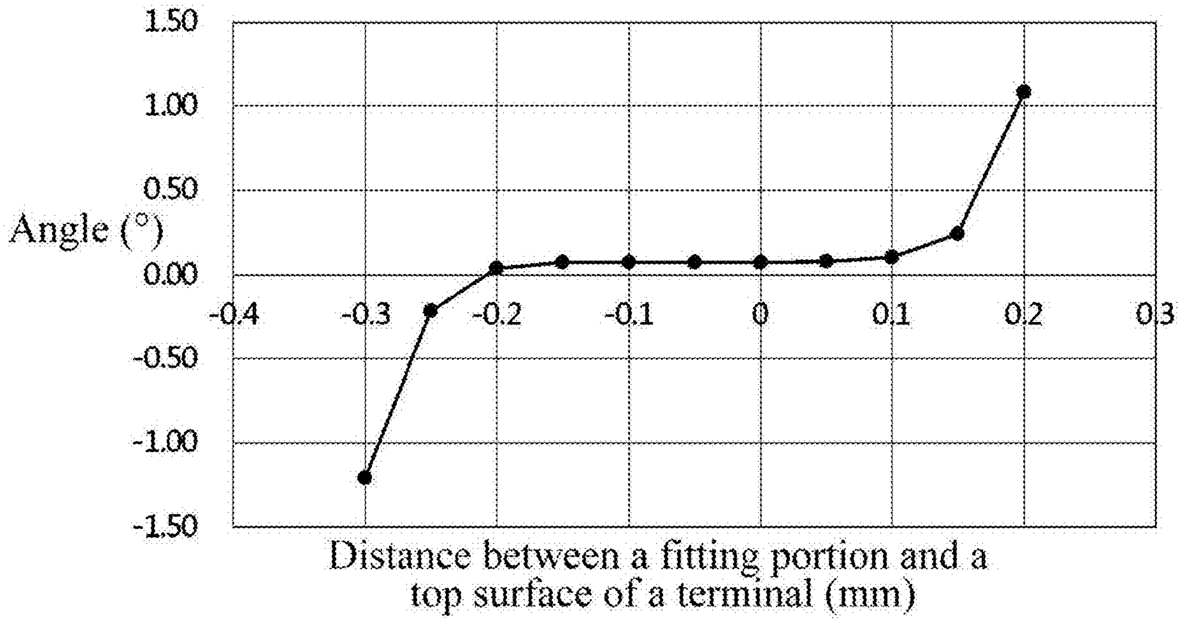

1.50
1.00
0.50
Angle (°)
0.00
-0.50
-1.00
-1.50
-0.4
-0.3
-0.2
-0.1
0
0.1
0.2
0.3
Distance between a fitting portion and a top surface of a terminal (mm)

FIG. 14

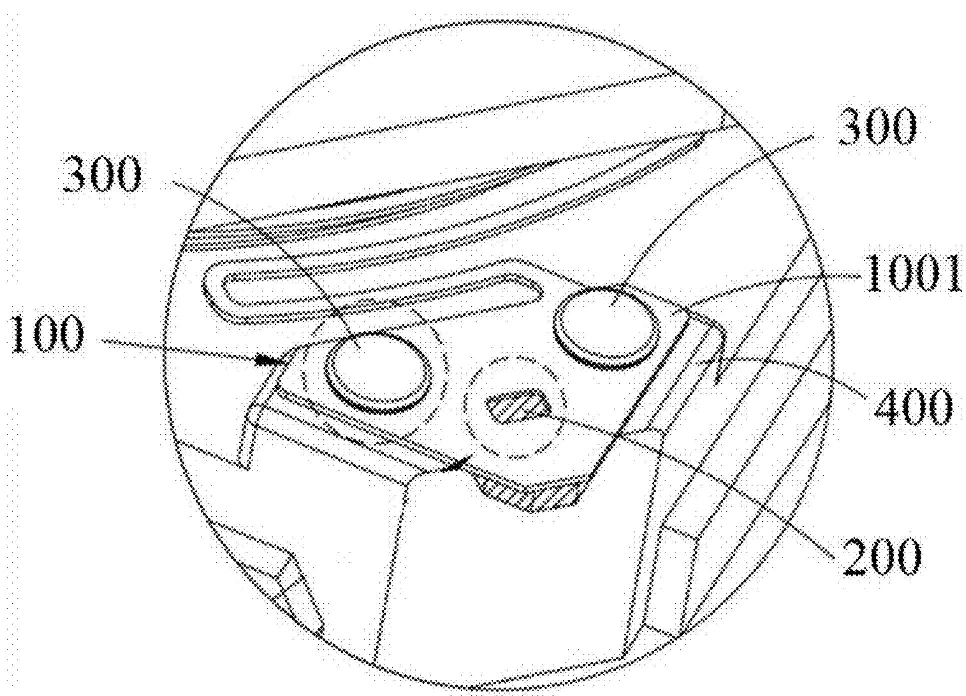

300
300
1001
100
400
200

FIG. 15

RESILIENT SHEET STRUCTURE AND VOICE COIL MOTOR

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims priority to Chinese Patent Application No. 202410207833.9 filed Feb. 26, 2024, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present invention relates to the field of voice coil motor technology and, in particular, to a resilient sheet structure and a voice coil motor.

BACKGROUND

[0003] A voice coil motor is an apparatus for converting electrical energy into mechanical energy and may implement a straight-line motion and a motion with a limited swing angle. The voice coil motor uses the interaction between the magnetic poles in the magnetic field from a permanent magnet and the magnetic field generated by an energized coil conductor to produce a regular motion.

[0004] The structure of the existing voice coil motor generally includes a carrier, and a channel is disposed in the carrier and configured to assemble the lens of a camera. A coil is disposed around the periphery of the carrier, and the corresponding coil is provided with a magnet. Thus, when the coil is energized, the carrier generates displacement in the same direction as electromagnetic induction force due to electromagnetic induction, thereby implementing zooming and focusing of the camera. At least one of the upper end surface and the lower end surface of the carrier is also fixedly connected to a resilient sheet. The resilient sheet is elastic and may deform under the action of the electromagnetic induction force. When the coil is not energized, the carrier can be restored to the initial state under the elastic force of the resilient sheet.

[0005] In the related art, as shown in FIG. **15**, the resilient sheet **100** generally includes a resilient sheet corner **1001** and a connection portion. The resilient sheet corner **1001** is connected to the resilient sheet connection region of a base **400**. The connection portion is connected to the lens located inside the voice coil motor to enable the resilient sheet to stabilize a moving lens through the connection portion. The resilient sheet **100** is usually fixed on the resilient sheet connection region by fusion and riveting. More specifically, the resilient sheet **100** is connected to a terminal **200** protruding from the resilient sheet connection region by fusion and is fixed on the resilient sheet connection region through a riveting member **300**. To improve the connection effect between the resilient sheet **100** and the terminal **200**, the riveting member **300** needs to be arranged on two sides of the connection part between the resilient sheet **100** and the terminal **200**, reducing the degree of freedom of design. The riveting position on the resilient sheet connection region is fixed, so that the riveting position cannot be adjusted according to the height of the terminal **200**. When the height of the terminal **200** is relatively high, a problem that the riveting position of the resilient sheet **100** cannot correspond to the riveting position on the resilient sheet connection region may occur, which may affect the fixed effect of the resilient sheet **100** and easily cause instability. As a result, the stable electrical connection between the resilient sheet **100** and the terminal **200** is affected, and the normal use of the voice coil motor is affected.

SUMMARY

[0006] The object of the present invention is to provide a resilient sheet structure and a voice coil motor. The resilient sheet structure can be suitable for flush-fitting with terminals of different heights, which has high connection stability and reliability.

[0007] As conceived previously, the following describes the solution of the present invention.

[0008] The resilient sheet structure includes a connection piece and a resilient sheet.

[0009] The resilient sheet includes a fixed portion, a beam, and a fitting portion. The fixed portion is electrically connected to the connection piece and is configured to be connected to the base assembly of the voice coil motor. One end of the beam is connected to the fixed portion, and the other end of the beam can rotate around one end of the beam, so that the beam is at an included angle with a preset plane. The fitting portion is connected to the other end of the beam, and the fitting portion can rotate around the beam, so that the fitting portion is parallel to the preset plane, and the fitting portion can be in flat contact with and be electrically connected to the top surface of the terminal on the base assembly.

[0010] The voice coil motor includes the resilient sheet structure as described above, the base assembly, and a coil assembly.

[0011] The base assembly includes a base, a connector, and a terminal. The base has a resilient sheet connection region. The connector is configured to fix the fixed portion of the resilient sheet structure to the resilient sheet connection region. The terminal is disposed on the base, and an end of the terminal extends to the resilient sheet connection region. The fitting portion of the resilient sheet structure is electrically connected to the terminal.

[0012] The coil assembly is disposed on the base assembly. The coil assembly includes a magnetic member disposed on the base assembly, a wire post electrically connected to the connection piece, a frame disposed on the base and formed with an annular groove, and a coil disposed on the annular groove and electrically connected to the wire post.

BRIEF DESCRIPTION OF DRAWINGS

[0013] FIG. **1** is view one illustrating the structure of a voice coil motor according to an embodiment of the present invention.

[0014] FIG. **2** is exploded view one of a voice coil motor according to an embodiment of the present invention.

[0015] FIG. **3** is exploded view two of a voice coil motor according to an embodiment of the present invention.

[0016] FIG. **4** is a view of a base and a resilient sheet structure according to an embodiment of the present invention.

[0017] FIG. **5** is a view of a resilient sheet structure according to an embodiment of the present invention.

[0018] FIG. **6** is an assembly view of a resilient sheet structure and a terminal according to an embodiment of the present invention.

[0019] FIG. **7** is a view illustrating the structure of a resilient sheet according to an embodiment of the present invention.

[0020] FIG. **8** is a top view of a resilient sheet structure according to an embodiment of the present invention.

[0021] FIG. **9** is a top view of a resilient sheet structure according to another embodiment of the present invention.

[0022] FIG. **10** is a top view of a resilient sheet structure according to yet another embodiment of the present invention.

[0023] FIG. **11** is a top view of a resilient sheet structure according to still yet another embodiment of the present invention.

[0024] FIG. **12** is a top view of a resilient sheet structure according to even still yet another embodiment of the present invention.

[0025] FIG. **13** is a size mark view of a resilient sheet according to an embodiment of the present invention.

[0026] FIG. **14** is a diagram showing the variation of the angle at which the other end of a beam rotates around one end of the beam with the distance between a fitting portion and the top surface of a terminal according to an embodiment of the present invention.

[0027] FIG. **15** is a view illustrating the structure of part of a voice coil motor in the related art according to an embodiment of the present invention.

REFERENCE LIST

[0028] **10** resilient sheet structure
[0029] **1** connection piece
[0030] **2** resilient sheet
[0031] **21** fixed portion
[0032] **211** riveting hole
[0033] **22** beam
[0034] **23** fitting portion
[0035] **231** welding hole
[0036] **24** torsion bar portion
[0037] **3** connection section
[0038] **20** base assembly
[0039] **201** base
[0040] **2011** resilient sheet connection region
[0041] **202** connector
[0042] **203** terminal
[0043] **2031** connection surface
[0044] **2032** welding object
[0045] **30** coil assembly
[0046] **40** outer housing
[0047] X first direction
[0048] Y second direction
[0049] Z third direction
[0050] **100** resilient sheet
[0051] **1001** resilient sheet corner
[0052] **200** terminal
[0053] **300** riveting member
[0054] **400** base

DETAILED DESCRIPTION

[0055] It is to be noted that similar reference numerals and letters represent similar items in the drawings. Therefore, once an item is defined in one drawing, the item no longer needs to be defined and interpreted in the subsequent drawings.

[0056] In the description of the present invention, unless otherwise expressly specified and limited, the term “connected to each other”, “connected” or “secured” is to be construed in a broad sense, for example, as securely connected, detachably connected or integrated; mechanically connected or electrically connected; directly connected to each other or indirectly connected to each other via an intermediary; or internal connection between two components or interaction relations between two components. For those of ordinary skill in the art, specific meanings of the preceding terms in the present invention may be construed based on specific situations.

[0057] In the present invention, unless otherwise expressly specified and limited, when a first feature is described as “above” or “below” a second feature, the first feature and the second feature may be in direct contact or be in contact via another feature between the two features. Moreover, when the first feature is described as “on”, “above”, or “over” the second feature, the first feature is right on, above, or over the second feature, the first feature is obliquely on, above, or over the second feature, or the first feature is simply at a higher level than the second feature. When the first feature is described as “under”, “below” or “underneath” the second feature, the first feature is right under, below or underneath the second feature, the first feature is obliquely under, below or underneath the second feature, or the first feature is simply at a lower level than the second feature. In the description of the embodiments, unless otherwise specified, “multiple” specifically refers to two or more.

[0058] In the description of the embodiments, the orientation or position relationships indicated by terms “above”, “below”, “right” and the like are based on the orientation or position relationships shown in the drawings. These orientations or position relationships are merely for ease of description and simplifying an operation and do not indicate or imply that the referred device or element has a specific orientation and is constructed and operated in a specific orientation. Thus, these orientations or position relationships are not to be construed as limiting the present invention. In addition, the terms “first” and “second” are used only for distinguishing between descriptions and have no special meaning.

[0059] An embodiment provides a voice coil motor having high connection stability and reliability.

[0060] As shown in FIGS. **1** to **4**, the voice coil motor includes a base assembly **20**, a coil assembly **30**, and a resilient sheet structure **10**. The base assembly **20** includes a base **201**, a connector **202**, and a terminal **203**. The base **201** has a resilient sheet connection region **2011**. The resilient sheet connection region **2011** may be disposed on the top of the base **201**. The connector **202** is configured to fix the resilient sheet structure **10** on the resilient sheet connection region **2011**, so that the resilient sheet structure **10** is connected to the base assembly **20**. The terminal **203** is disposed on the base **201**. Specifically, one end of the terminal **203** extends to the resilient sheet connection region **2011**, and the other end of the terminal **203** protrudes from the bottom of the base **201** to be electrically connected to other electronic devices.

[0061] The resilient sheet structure **10** is electrically connected to the part of the terminal **203** located in the resilient sheet connection region **2011**. In some optional embodiments, the terminal **203** has a connection surface **2031** exposed from the resilient sheet connection region **2011**. The resilient sheet structure **10** is electrically connected to the connection surface **2031**. For example, the connection method between the resilient sheet structure **10** and the terminal **203** may be laser welding, so that the connection effect is better. When the resilient sheet structure **10** is connected to the terminal **203**, as shown in FIG. **6**, a welding object **2032** may be disposed between the resilient sheet structure **10** and the terminal **203**, and the welding object **2032** may be fused so that the resilient sheet structure **10** is connected to the terminal **203**.

[0062] The coil assembly **30** is disposed on the base assembly **20**. The coil assembly **30** includes a magnetic member disposed on the base assembly **20**, a wire post electrically connected to the resilient sheet structure **10**, a frame disposed on the base **201** and formed with an annular groove, and a coil disposed on the annular groove and electrically connected to the wire post. In some optional embodiments, the voice coil motor also includes an upper resilient sheet. The upper resilient sheet is disposed on the frame.

[0063] In some embodiments, the voice coil motor also includes an outer housing **40**. The outer housing **40** covers at least part of the base assembly **20** and the coil assembly **30**. The outer housing **40** is configured to protect the base assembly **20** and the coil assembly **30** to prevent moisture or dust from entering each component and causing a decrease in the service life of an element.

[0064] In some embodiments, the terminal **203** may also have an input/output end. The coil, the wire post, the resilient sheet structure **10**, and the terminal **203** form a circuit with a power supply through the input/output end to match the magnetic member to move the lens (not shown) located at the center of the voice coil motor. In other words, a current enters the coil from the input/output end of the terminal **203** along the resilient sheet structure **10** and the wire post to form an electric field. The magnetic field generated by the preceding electric field may interact with the magnetic member and move the lens located at the center of the voice coil motor.

[0065] This embodiment may describe the resilient sheet structure **10** in detail below.

[0066] As shown in FIGS. **5** to **13**, the resilient sheet structure **10** includes a connection piece **1** and a resilient sheet **2**. The connection piece **1** is configured to be electrically connected to the coil. In some optional embodiments, as shown in FIG. **5**, the connection piece **1** may be fixedly and electrically connected to the resilient sheet **2** through a connection section **3**. In this embodiment, the resilient sheet **2** and the connection piece **1** are sheet-shaped.

[0067] Referring to FIG. **5**, the resilient sheet **2** includes a fixed portion **21**, a beam **22**, and a fitting portion **23**. The fixed portion **21** is fixed and electrically connected to the connection piece **1**, and the fixed portion **21** is configured to be connected to the resilient sheet connection region **2011** of the base **201**, that is, the fixed portion **21** is fixed on the resilient sheet connection region **2011** through the connector **202**. In some optional embodiments, the fixed portion **21** is formed with a riveting hole **211**. The fixed portion **21** may be fixed on the resilient sheet connection region **2011** by a riveting and pressing process.

[0068] One end of the beam **22** is connected to the fixed portion **21**, and the fitting portion **23** is connected to the other end of the beam **22**. The other end of the beam **22** can rotate around one end of the beam **22**, that is, the other end of the beam **22** can rotate with one end of the beam **22** as a fulcrum, so that the beam **22** is at an included angle with a preset plane. The preset plane may overlap or be parallel to the connection surface **2031**. For example, as shown in FIG. **7**, when the beam **22** rotates to an inclined state, the included angle between the beam **22** and the preset plane is c1. It is to be noted that when the other end of the beam **22** rotates around one end of the beam **22**, the fitting portion **23** can be driven to rotate together, that is, the fitting portion **23** can rotate relative to one end of the beam **22**.

[0069] In this embodiment, the fitting portion **23** can rotate around the beam **22**, that is, the fitting portion **23** can rotate with the beam **22** as a fulcrum or a rotation shaft. The rotation direction of the other end of the beam **22** relative to one end of the beam **22** is opposite to the rotation direction of the fitting portion **23** relative to the beam **22**, so that the fitting portion **23** is parallel to the preset plane. Thus, the fitting portion **23** can be in flat contact with and electrically connected to the connection surface **2031** of the terminal **203** on the base assembly **20**. For example, after the fitting portion **23** rotates around the beam **22**, the included angle between the fitting portion **23** and the beam **22** is c2. When c1=c2, that is, when the angle at which the other end of the beam **22** rotates around one end of the beam **22** is equal to the angle at which the fitting portion **23** rotates around the beam **22**, the fitting portion **23** is parallel to the preset plane, and the preset plane is parallel to or overlaps the connection surface **2031**, so that the fitting portion **23** can be in flat contact with the connection surface **2031**. The fitting portion **23** is plate-like to better contact with the terminal **203**. The fitting portion **23** is connected to the terminal **203** in a laser fusion manner, so that the connection reliability is relatively high. For example, as shown in FIG. **9**, the fitting portion **23** is planar-ring shaped and is formed with a welding hole **231**. The welding hole **231** is configured to place a welding object **2032** between the fitting portion **23** and the terminal **203**.

[0070] When the resilient sheet structure **10** provided in this embodiment is used, the connection piece **1** may be electrically connected to the coil. Then, the fixed portion **21** is connected to the resilient sheet connection region **2011** of the base **201**. Alternatively, the fixed portion **21** may also be connected to the resilient sheet connection region **2011** of the base **201**. Then, the connection piece **1** may be electrically connected to the coil. This is not limited in this embodiment. At this time, the fixed portion **21**, the beam **22**, and the fitting portion **23** are coplanar and in a horizontal state. For example, the fixed portion **21**, the beam **22**, and the fitting portion **23** are in contact with the resilient sheet connection region **2011** of the base **201**. Alternatively, when the terminal **203** protrudes from the resilient sheet connection region **2011**, that is, when the height of the terminal **203** is relatively high, the fitting portion **23** may be in a tilted state. Specifically, the fitting portion **23** tilts up in a second direction Y shown in FIG. **5**. Next, the other end of the beam **22** is controlled to rotate around one end of the beam **22**. At this time, the other end of the beam **22** drives the fitting portion **23** to move in a direction away from the base **201** until there is an included angle between the beam **22** and the top surface of the base **201**. Then, the fitting portion **23** is controlled to rotate around the other end of the beam **22**, so that there is an included angle between the fitting portion **23** and the beam **22** until the fitting portion **23** can fully contact with the connection surface **2031** of the terminal **203**, that is, the fitting portion **23** is parallel to the preset plane. Then, laser welding is performed between the fitting portion **23** and the terminal **203**.

[0071] The resilient sheet structure **10** is provided in this embodiment, the resilient sheet **2** includes a fixed portion **21**, a beam **22**, and a fitting portion **23**. The fixed portion **21** is connected to the resilient sheet connection region **2011** of the base **201**. One end of the beam **22** is connected to the fixed portion **21**, and the other end of the beam **22** is connected to the fitting portion **23**. The other end of the beam **22** can rotate around one end of the beam **22**, that is, the beam **22** rotates from a state in which the beam **22** is parallel to the preset plane to an inclined state that the beam **22** is at an included angle with the preset plane. That is, the other end of the beam **22** can be tilted relative to the fixed portion **21**. Specifically, the beam **22** is tilted relative to the fixed portion **21** in the second Y direction shown in FIG. **5**. The fitting portion **23** is disposed at the other end of the beam **22**. When the other end of the beam **22** rotates around the one end of the beam **22**, the fitting portion **23** may also move along with the other end of the beam **22**, and the fitting portion **23** can rotate independently around the other end of the beam **22**. Since the tilted height of the beam **22** relative to the base **201** and the angle of the fitting portion **23** around the other end of the beam **22** are adjustable, the distance between the fitting portion **23** and the base **201** is adjusted, which is applicable to terminals **203** of different heights. Thus, the flexibility of the fitting portion **23** is improved, and it is convenient for the fitting portion **23** to be parallel to the connection surface **2031**, so that the fitting portion **23** can be flush-fitting with the connection surface **2031**, and the fitting portion **23** and the connection surface **2031** have a relatively large contact area. Moreover, the connection stability and reliability between the fitting portion **23** and the terminal **203** are improved, and the normal use of the voice coil motor that applies the resilient sheet structure **10** is ensured.

[0072] Moreover, since the fitting portion **23** has a degree of freedom, the number and the position of the riveting and pressing on the fixed portion **21** are not limited.

[0073] It is to be noted that to improve the integrity of the resilient sheet **2**, the resilient sheet **2** is a one-piece construction. Furthermore, the resilient sheet structure **10** is a one-piece construction, that is, the resilient sheet structure **10** can be formed by stamping and cutting a plate.

[0074] As shown in FIGS. **7** and **8**, the resilient sheet **2** also includes a torsion bar portion **24**. The other end of the beam **22** is connected to the torsion bar portion **24**. The fitting portion **23** is connected to the torsion bar portion **24**, that is, the other end of the beam **22** is connected to the fitting portion **23** through the torsion bar portion **24**. The fitting portion **23** can rotate around the torsion bar portion **24**. With the arrangement of the bar portion **24**, the flexibility of the resilient sheet structure **10** can be further improved and the torsional stress between the fitting portion **23** and the beam **22** can also be adjusted.

[0075] In some optional embodiments, the torsion bar portion **24** can rotate around the beam **22**. The rotation axis of the torsion bar portion **24** is parallel to the rotation axis of the beam **22**. The torsion bar **24** can rotate around the beam **22** to further adjust the distance between the fitting portion **23** and the top surface of the base **201**, so that the resilient sheet structure **10** is more flexible. In some optional embodiments, the extension direction of the torsion bar portion **24** is perpendicular to the extension direction of the beam **22**. In other optional embodiments, the extension direction of the torsion bar portion **24** may be at an included angle with the extension direction of the beam **22**. In this embodiment, the torsion bar portion **24** extends in a first direction X.

[0076] For example, the beam **22**, the torsion bar portion **24**, and the fitting portion **23** are all located on the same side of the fixed portion **21**. More specifically, the beam **22**, the torsion bar portion **24**, and the fitting portion **23** are all located on a side of the fixed portion **21** in a third direction Z, it is convenient that the beam **22**, the torsion bar portion **24**, and the fitting portion **23** are tilted relative to the fixed portion **21**, and the fitting portion **23** is connected to the terminal **203**.

[0077] There may be one or more beams **22**. This is not limited in this embodiment.

[0078] In some optional embodiments, as shown in FIGS. **11** and **12**, one beam **22** is provided. The fitting portion **23** is located on a side of the beam **22** and is spaced apart from the middle portion of the beam **22**.

[0079] In other optional embodiments, as shown in FIGS. **9** and **10**, two beams **22** are provided. The two beams **22** are disposed at intervals in the first direction X. The end of each of two beams **22** is connected to the fixed portion **21**, and the other ends of the two beams **22** are connected to the fitting portion **23**. With the arrangement of the two beams **22**, the reliability for connecting the fitting portion **23** is improved.

[0080] Continuously referring to FIGS. **9** and **10**, when two beams **22** are provided, the fitting portion **23** is located between the two beams **22** in the first direction X. In other embodiments, the fitting portion **23** may also be located outside the two beams **22** instead of between the two beams **22**. This is not limited in this embodiment.

[0081] The beam **22** having the preceding two structures can be tilted relative to the fixed portion **21**, so that there is a certain distance between the fitting portion **23** and the base **201**.

[0082] In this embodiment, the beam **22** is any one or a combination of a straight-line structure, a curved structure, or a bent-line structure, which may be specifically selected or designed according to an application position or other scenarios. This is not limited in this embodiment. The beam **22** in FIG. **8** is a straight-line structure. The beams **22** in FIG. **9** and FIG. **10** are bent-line structures and specifically include two line segments. The difference is that the bending directions of the two line segments are different. The beams **22** in FIG. **11** and FIG. **12** are combinations of a bent-line structure and a curved structure.

[0083] FIG. **13** is a view of part of the resilient sheet structure **10** according to this embodiment, and in FIG. **13**, the beam **22**, the fitting portion **23**, and the torsion bar portion **24** are marked with sizes.

[0084] A calculation is performed according to the material mechanics. The rotation angle formulas of the beam **22**, the torsion bar portion **24**, and the fitting portion **23** are listed, so that the position where the rotation angle is zero (that is, the fitting portion **23** has no inclination) may be obtained. This position may also be obtained by analog simulation.

[0085] The rotation angle formula of the beam **22** includes two parts, one part is

$$\theta_{max} = \frac{FL^2}{2EI}$$

and the other part is

$$\theta_{max} = \frac{M_0 L}{EI}.$$

The rotation angle formula of the torsion bar portion **24** is:

$$\Delta\phi = \frac{M_0 L}{GJ}.$$

The rotation angle formula of the fitting portion **23** is:

$$\theta = -\frac{Fa^2}{2EI}.$$

[0086] According to the material mechanics theory, the derivation is available below.

$$\frac{FL^2}{2EI_1} - \frac{M_0 L_1}{EI_1} - \frac{M_0 L_2}{GJ_2} - \frac{Fa^2}{2EI_3} = 0 \begin{cases} M_0 = F \cdot a \\ G = \dfrac{E}{2(1+v)} \end{cases}$$

$$\frac{1}{I_3}a^2 + \left(\frac{2L_1}{I_1} - \frac{4(1+v)L_2}{J_2}\right)a - \frac{L^2}{I_1} = 0$$

$$a = \frac{-\left(\frac{2L_1}{I_1} - \frac{4(1+v)L_2}{J_2}\right) \pm \sqrt{\left(\frac{2L_1}{I_1} - \frac{4(1+v)L_2}{J_2}\right)^2 - 4\frac{1}{I_3}\left(-\frac{L^2}{I_1}\right)}}{2\frac{1}{I_3}}$$

[0087] In the preceding derivation process, F denotes the force on a component. M_0 denotes torque. E, I, v, and J are constants in the material mechanics. E denotes a tensile modulus of elasticity. I denotes a moment of inertia. G denotes a shear modulus of elasticity. J denotes compliance. v denotes a Poisson's ratio. In subscripts, 1 denotes the parameter corresponding to the beam **22**, 2 denotes the parameter corresponding to the torsion bar portion **24**, and 3 denotes the parameter corresponding to the fitting portion **23**. As shown in FIG. **13**, the obtained position where a rotation angle is 0 is represented by a dashed line passing through the welding hole.

[0088] FIG. **14** is a line chart where the distance between the fitting portion **23** and the top surface (that is, the connection surface **2031**) of the terminal **203** corresponds to the angle at which the other end of the beam **22** rotates around one end of the beam **22** according to this embodiment. It can be seen that as long as the terminal **203** covers a non-inclined position, the fitting portion **23** of the resilient sheet structure **10** may always remain in flat contact with the connection surface **2031** of the terminal **203**.

[0089] The resilient sheet structure **10** provided in this embodiment uses geometric design to implement an adaptive structure. There is no limit to the rigidity of the resilient sheet structure **10**. Regardless of whether the rigidity is large or small, the fitting portion **23** of the resilient sheet structure **10** may be in flat contact with the connection surface **2031** of the terminal **203**. The protrusion height of the terminal **203** does not affect the gap between the fitting portion **23** of the resilient sheet structure **10** and the connection surface **2031**, so that the fitting portion **23** is completely in flat contact with the surface of the terminal **203** without any gap, thereby increasing the robustness of the manufacturing process of the product.

What is claimed is:

1. A resilient sheet structure, comprising:

a connection piece; and

a resilient sheet comprising a fixed portion, a beam, and a fitting portion, wherein the fixed portion is electrically connected to the connection piece and is configured to be connected to a base assembly of a voice coil motor; one end of the beam is connected to the fixed portion, and another end of the beam is capable of rotating around the one end of the beam to enable the beam to be at an included angle with a preset plane; and the fitting portion is connected to the another end of the beam, and the fitting portion is capable of rotating around the beam to enable the fitting portion to be parallel to the preset plane and to enable the fitting portion to be in contact with and to be electrically connected to a top surface of a terminal on the base assembly.

2. The resilient sheet structure according to claim **1**, wherein the resilient sheet further comprises a torsion bar portion, the another end of the beam is connected to the torsion bar portion, the fitting portion is connected to the torsion bar portion, the torsion bar portion is capable of rotating around the beam, and the fitting portion is capable of rotating around the torsion bar portion.

3. The resilient sheet structure according to claim **1**, wherein one beam is provided, and the fitting portion is located on a side of the beam.

4. The resilient sheet structure according to claim **1**, wherein two beams are provided, the two beams are disposed at intervals in a first direction, one end of each of the two beams is connected to the fixed portion, and another end of each of the two beams is connected to the fitting portion.

5. The resilient sheet structure according to claim **4**, wherein the fitting portion is located between the two beams in the first direction.

6. The resilient sheet structure according to claim **1**, wherein the beam is any one or a combination of a straight-line structure, a curved structure, or a bent-line structure.

7. The resilient sheet structure according to claim **2**, wherein the beam is any one or a combination of a straight-line structure, a curved structure, or a bent-line structure.

8. The resilient sheet structure according to claim **3**, wherein the beam is any one or a combination of a straight-line structure, a curved structure, or a bent-line structure.

9. The resilient sheet structure according to claim **1**, wherein the resilient sheet is a one-piece construction.

10. The resilient sheet structure according to claim **1**, wherein the beam and the fitting portion are located on a same side of the fixed portion.

11. The resilient sheet structure according to claim **1**, wherein the fitting portion is planar-ring shaped and the fitting portion is formed with a welding hole.

12. A voice coil motor, comprising:

the resilient sheet structure;

the base assembly comprising a base, a connector, and the terminal, wherein the base has a resilient sheet connection region, the connector is configured to fix the fixed portion of the resilient sheet structure to the resilient sheet connection region, the terminal is disposed on the base, an end of the terminal extends to the resilient sheet connection region, and the fitting portion of the resilient sheet structure is electrically connected to the terminal; and

a coil assembly disposed on the base assembly, comprising a magnetic member disposed on the base assembly, a wire post electrically connected to the connection piece, a frame disposed on the base and formed with an annular groove, and a coil disposed in the annular groove and electrically connected to the wire post;

wherein the resilient sheet structure, comprising:

a connection piece; and

a resilient sheet comprising a fixed portion, a beam, and a fitting portion, wherein the fixed portion is electrically connected to the connection piece and is configured to be connected to a base assembly of a voice coil motor; one end of the beam is connected to the fixed portion, and another end of the beam is capable of rotating around the one end of the beam to enable the beam to be at an included angle with a preset plane; and the fitting portion is connected to the another end of the beam, and the fitting portion is capable of rotating around the beam to enable the fitting portion to be parallel to the preset plane, and to enable the fitting portion fit to and being electrically connected to a top surface of a terminal on the base assembly.

13. The voice coil motor according to claim **12**, wherein the resilient sheet further comprises a torsion bar portion, the another end of the beam is connected to the torsion bar portion, the fitting portion is connected to the torsion bar portion, the torsion bar portion is capable of rotating around the beam, and the fitting portion is capable of rotating around the torsion bar portion.

14. The voice coil motor according to claim **12**, wherein one beam is provided, and the fitting portion is located on a side of the beam.

15. The voice coil motor according to claim **12**, wherein two beams are provided, the two beams are disposed at intervals in a first direction, one end of each of the two beams is connected to the fixed portion, and another end of each of the two beams is connected to the fitting portion.

16. The voice coil motor according to claim **15**, wherein the fitting portion is located between the two beams in the first direction.

17. The voice coil motor according to claim **12**, wherein the beam is any one or a combination of a straight-line structure, a curved structure, or a bent-line structure.

18. The voice coil motor according to claim **12**, wherein the resilient sheet is a one-piece construction.

19. The voice coil motor according to claim **12**, wherein the beam and the fitting portion are located on a same side of the fixed portion.

20. The voice coil motor according to claim **12**, wherein the fitting portion is planar-ring shaped and the fitting portion is formed with a welding hole.

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